

Closing Sim2Real Gap via Bi-level Reinforcement Learning

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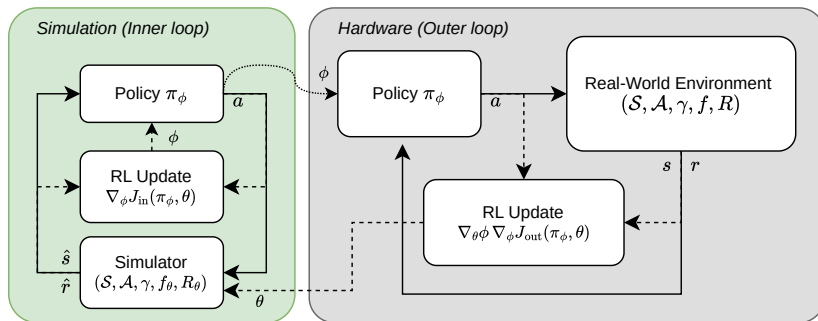
Outline

1 RL-over-RL Motivation and Bi-Level Formulation

Closing Sim2Real gap via Bi-level RL

- Sim2Real Gap:
 - ▶ Improving simulator accuracy and robustness.
 - ▶ Accuracy and robustness does not guarantee optimal real-world performance.
 - ▶ Optimize simulator parameters θ to maximize real-world return.

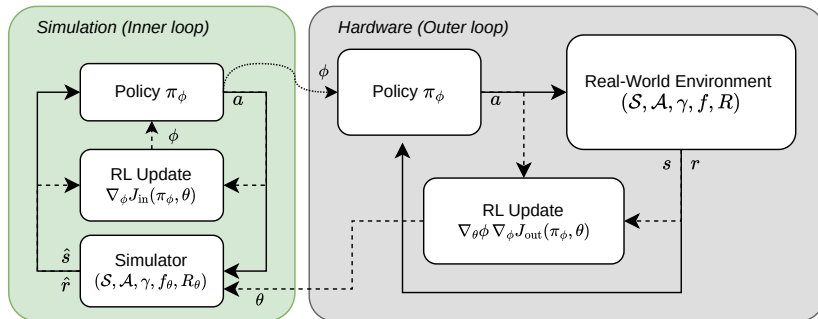
Closing Sim2Real gap via Bi-level RL



This leads to a bi-level optimization problem:

$$\max_{\theta} J(\pi_{\phi(\theta)}, \text{real world}) \quad \text{s.t.} \quad \phi(\theta) = \arg \max_{\phi} \hat{J}(\pi_{\phi}, f_{\theta}, R_{\theta}).$$

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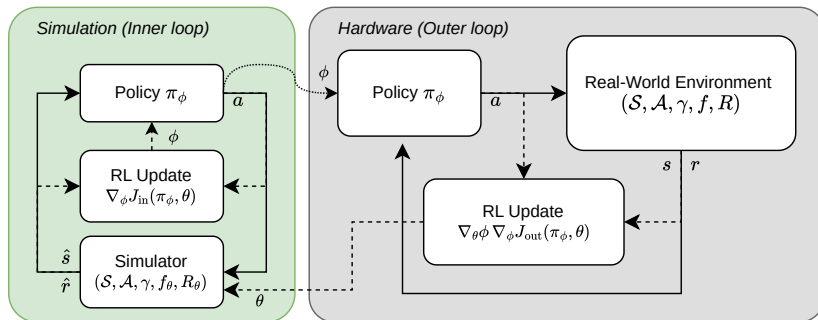


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- **Inner level:** In-sim RL (SPG) finds π_ϕ optimal for simulator.
- **Outer level:** Updates simulator parameters θ to improve real-world performance.

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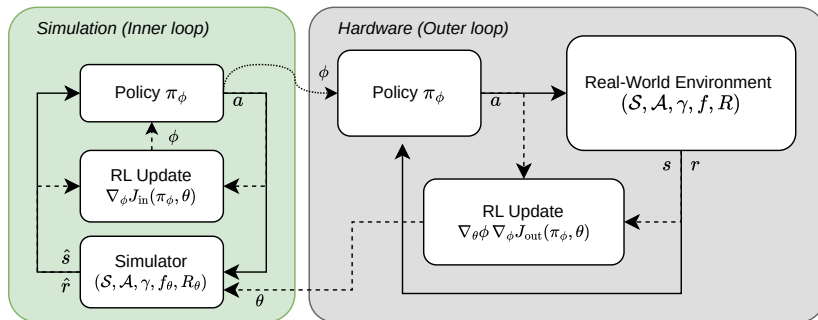
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$$\nabla_{\theta} J_{\text{real}}(\pi_{\phi(\theta)}) = \underbrace{\nabla_{\theta} \phi(\theta)}_{\text{policy sensitivity}} \underbrace{\mathbb{E}_{s \sim \rho^{\pi_{\phi}}, a \sim \pi_{\phi}(\cdot|s)} [\nabla_{\phi} \log \pi_{\phi}(a|s) Q^{\pi_{\phi}}(s, a)]}_{\text{policy gradient (real world)}}.$$

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$\nabla_{\theta} \phi$: To compute outer gradients, we need how the inner policy π_{ϕ} changes when simulator parameters θ are perturbed.

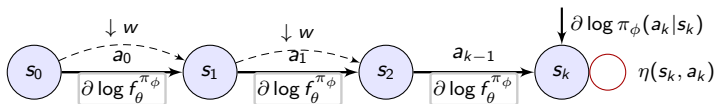
Sensitivity of In-Sim RL

- Inner SPG defines an implicit condition: $\hat{\phi}(\phi, \theta) = \mathbb{E}_{\hat{\rho}_{\theta}^{\pi_{\phi}}} [\nabla_{\phi} \log \pi_{\phi} \hat{Q}_{\theta}^{\pi_{\phi}}] = 0$.
- $\nabla_{\theta} \phi$ using the Implicit Function Theorem applied to this condition.
- Conceptually, this requires:
 - 1 Estimating the sensitivity of the markov chain of the simulator under a policy w.r.t. θ .
 - 2 Estimating the sensit of in-sim critic w.r.t simulator parameters θ .
 - 3 Combining these to obtain the sensitivity of in-sim policy: $\nabla_{\theta} \phi$
- In practice, these quantities can be estimated using in-sim rollouts (no need for real-world exploration).
- Combine real-world PG with the in-sim sensitivities to update θ for better real return.

Differentiating the Markov Chain

Sensitivity of a Markov chain expectation: Differentiate an expectation at a fixed step k under the in-sim Markov chain.

$$\partial \mathbb{E}_{s \sim \hat{\rho}_{\theta, k}^{\pi_{\phi}}} [\eta(s_k, a_k)] = \mathbb{E}_{\substack{s_0 \sim \rho^0, a_i \sim \pi_{\phi}, \\ s_{i+1} \sim f_{\theta}^{\pi_{\phi}}}} \left[\eta(s_k, a_k) \left(\partial \log \pi_{\phi}(a_k | s_k) + \sum_{i=0}^{k-1} \partial \log f_{\theta}^{\pi_{\phi}}(s_{i+1} | s_i, a_i) \right) \right]$$



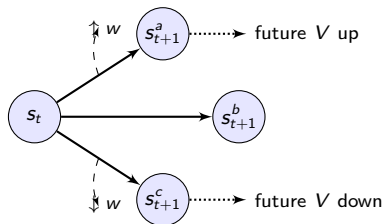
- The sensitivity estimation amounts to reweighting each trajectory by how its likelihood changes under ∂ and adjusting at step k for the chosen action.

Differentiating the Critic

The critic satisfies a fixed point under the bellman operator; change in θ perturbs every backup along the trajectory tree.

$$\begin{aligned}\partial Q^\pi(s, a) &= \partial R(s, a) + \gamma \mathbb{E}_{s_+ \sim f} [\partial V^\pi(s_+) + V^\pi(s_+) \partial \log f(s_+ | s, a)], \\ \partial V^\pi(s) &= \mathbb{E}_{a \sim \pi} [\partial Q^\pi(s, a) + Q^\pi(s, a) \partial \log \pi(a | s)].\end{aligned}$$

- Start at $Q(s_t, a_t)$.
- A small $\delta\theta$
 - 1 perturbs $R_\theta(s_t, a_t)$ (local),
 - 2 re-weights next states via f_θ (global),
 - 3 and keeps propagating through $V(s_{t+1}), V(s_{t+2}), \dots$



$\delta\theta$ changes edge weights (transition probs), so the Bellman backup shifts via both reward and where we go next.

- Immediate effect: rewards $R_\theta(s, a)$ shift with θ .
- Propagated effect: transition shifts alter *who* we next visit $\Rightarrow$ future values change.
- The overall sensitivity is a *backward* accumulation along edges of the trajectory tree.